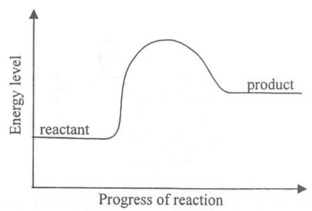
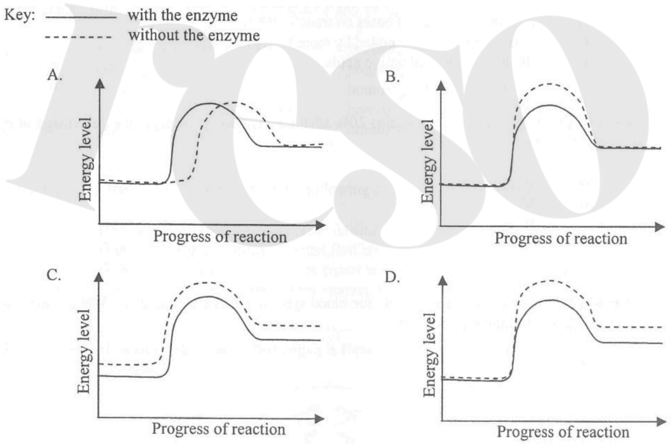
1. 2013Q6
Questions 6 and 7 refer to the graph below, which shows the energy levels of the reactant and product of a biochemical reaction in the presence of its enzymes:

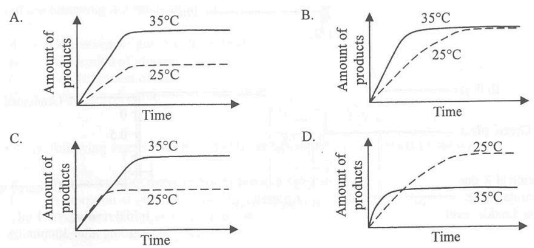


The reaction shown in the graph is
A. an anabolic process because energy is absorbed.
B. an anabolic process because energy is released.
C. a catabolic process because energy is absorbed.
D. a catabolic process because energy is released.

2. 2013Q7
Which of the following graphs correctly shows the change in the energy level of the reaction if the enzyme involved is absent?



3. 2014Q9
Which of the following graphs correctly shows the changes in the amount of products in a reaction catalysed by a human enzyme at different temperatures?



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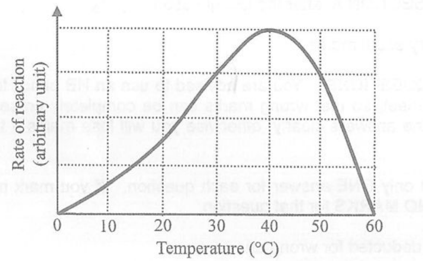
4. 2015Q3
The graph below shows the effect of temperature on enzyme activity:



Which of the following statements correctly describes the enzyme reaction?
A. The enzyme is denatured at 0°C and 60°C
B. The reaction taking place at 50°C is faster than that at 20°C
C. There are more collisions between substrate and enzyme molecules at 40°C than 60°C
D. The amount of product collected at the end of the reaction is greatest if the reaction takes place at 40°C

5. 2016Q27
Which of the following statements about enzymes is ***incorrect***?
A. Enzymes are produced by cells.
B. Enzymes are denatured at extreme temperatures.
C. There could be more than one specific enzyme to catalyse the same reaction.
D. When an enzyme encounters the same substrates, it always produces the same products.

6. 2017Q5
Different animals produce different maltases to digest maltose. The maltases produced have different molecular sizes. Which of the following descriptions of these maltases is correct?
A. Their active sites have similar shape.
B. Their amino acid sequences are the same.
C. They have the same optimum temperature.
D. They have the same three-dimensional structure.

7. 2018Q2
Questions 2 to 4 refer to an experiment on the enzyme catalase, which speeds up the breakdown of hydrogen peroxide to release oxygen. John added a 1 cm^3 cube of pig liver to a boiling tube containing 5 mL hydrogen peroxide solution. Gas bubbles were released and he used a glowing splint to test the gas. He repeated the experiment with beef, potato and apple. The results are shown below:

Tissue	***Rate of bubbles released***	***Glowing split re-lit***
Pig liver	Moderate	Yes
Beef	Moderate	Yes
Potato	Slow	Yes
Apple	Slow	Yes

Which of the following statements is an observation of the experiment?
A. These tissues contained catalase.
B. Oxygen gas was released in the reaction.
C. The gas released re-lit the glowing splint
D. Animal tissues had more catalase than plant tissues.

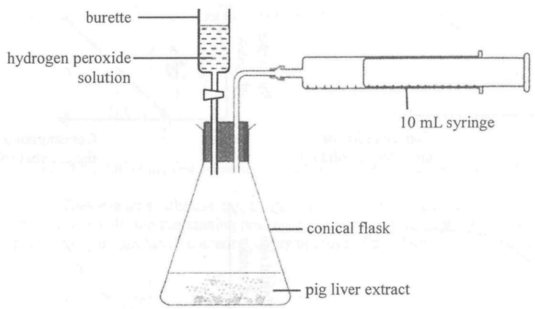
8. 2018Q3
When the release of gas bubbles had stopped, John added more hydrogen peroxide solution to the boiling tubes. Which of the following combinations correctly shows the expected result and the explanation of this additional experiment?

	Expected result	***Explanation***
A.	Gas bubbles released	Catalase is specific in its action.
B.	Gas bubbles released	Catalase is reusable.
C.	No gas bubbles	Catalase has been used up.
D.	No gas bubbles	Catalase is denatured.

9. 2018Q4
In order to prove that hydrogen peroxide is the substrate of this enzymatic reaction, which of the following steps should be used as a control?
A. Repeat the experiment using water and the tissues.
B. Repeat the experiment using water and boiled tissues.
C. Repeat the experiment using hydrogen peroxide but no tissues.
D. Repeat the experiment using hydrogen peroxide and boiled tissues.

10. 2018Q27
'Lock and Key" is a scientific model which is a selective representation used to explain that enzymes
A. are biological catalysts.
B. are specific in action.
C. are protein in nature.
D. are required in small amounts.

11. 2019Q9
Questions 9 and 10 refer to the diagram below, which shows an experimental set-up prepared by a student to investigate the effect of temperature on catalase activity. Pig liver extract contains catalase which speeds up the breakdown of hydrogen peroxide into oxygen and water. A fixed volume of hydrogen peroxide solution was added to the liver extract and a 10mL syringe was used to collect the oxygen gas released from the reaction mixture.



In the trial run conducted at room temperature, the student found that the volume of oxygen released was greater than the maximum collection volume of the syringe. How should he modify the set-up in order to collect valid data when repeating the experiment at different temperatures?
(1) use a larger syringe
(2) use a larger conical flask
(3) reduce the volume of the hydrogen peroxide solution added
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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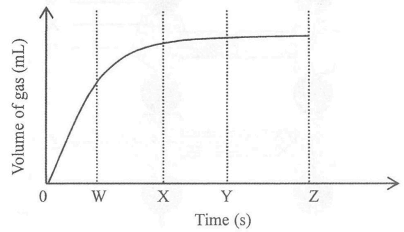
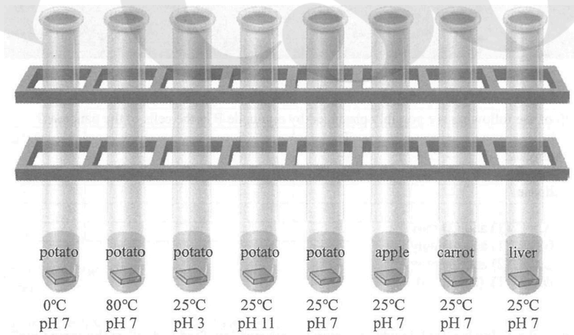
12. 2019Q10
After modifying the set-up, the following graph was obtained which shows the volume of gas collected over time at room temperature:



The student planned to use the volume of gas collected over a fixed period of time as the dependent variable to study the effect of different temperatures on catalase activity. Which of the following is the most suitable time period for the measurement?

A. 0 - W
B. 0 - X
C. 0 - Y
D. 0 - Z

13. 2020Q6
Questions 6 and 7 refer to the diagram below, which shows a set-up for investigating the activity of catalase in living tissues. Catalase is an enzyme which can break down hydrogen peroxide. Each test tube contains the same amount of hydrogen peroxide solution at the same concentration. A piece of living tissue is added into each tube as indicated below:



How many independent variables are being studied in this investigation?
A. 2
B. 3
C. 4
D. 8

14. 2020Q7
Which of the following control variables is most important for a fair comparison in the above investigation?
A. mass of living tissues
B. shape of living tissues
C. volume of living tissues
D. surface area of living tissues

15. 2020Q8
Which of the following descriptions of the active site of an enzyme is correct?
(1) It can be used again.
(2) It is the part of the enzyme on which its substrate can fit.
(3) Its shape is determined by the amino acid sequence of the enzyme.
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

16. 2023Q15
Questions 15 to 17 refer to an investigation about enzymes in fruits. Amy learned that some fruits contained proteases which can be used as meat tenderisers. She wanted to find a suitable fruit juice for the slow cooking of steak, i.e. mixing the steak with fruit juice in a sealed bag and then cooking it in a water bath set at around 50°C to 70°C for 1 hour. The procedure of her investigation is shown below:
1. Extract fruit juice from pineapple.
2. Add 15mL of pineapple juice in 3 boiling tubes respectively with labels 50°C, 60°C and 70°C.
3. Put the tubes into water baths set at respective temperatures for 1 hour.
4. Let the tubes cool down to room temperature.
5. Add 3 egg white cubes of size 1 cm^3 into each cube.
6. (Step for the measurement of the dependent variable)
7. Repeat the above steps with papaya, kiwi and lemon instead of pineapple.

Based on the procedure given, how many independent variable(s) is / are being studied?
A. 1
B. 2
C. 3
D. 4

17. 2023Q16
Step 6 is about the measurement of the dependent variable. Which of the following can be used for measuring the dependent variable for this investigation?
(1) the time taken for the disappearance of the egg white cubes
(2) the change in shape of the egg white cubes after a fixed time
(3) the change in weight of the egg white cubes after a fixed time
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

18. 2023Q17
Which of the following is the assumption for this investigation?
A. The proteases are present in the fruits tested.
B. The proteases remain active at high temperatures.
C. The proteases are still functioning after extraction.
D. The actions of the proteases on egg white cubes and steaks are similar.

LQ
1. 2013Q10 - Essay
Proteins serve different functions in our body and their functional role is clearly related to their conformation (shape). Describe how protein molecules can have different conformations and explain how the different conformations enable them to carry out different functions. (11 marks)

2. 2019Q2
The schematic diagram below shows a reaction mixture of an anabolic reaction catalysed by an enzyme. Drawings P, Q, R, S, and T represent different components of the mixture:

a) Which drawing represents the substrate in this anabolic reaction? Explain your answer. (2 marks)

b) Which drawing represents the enzyme? Explain your answer. (2 marks)

3. 2021Q4

Glycogen and a disaccharide named trehalose are two common types of energy reserve found in insect species A. An experiment was conducted to study the energy reserve used for flying in this insect species. Three groups of insect species A were respectively injected with equal volumes of physiological saline, an inhibitor of trehalose-digesting enzymes and an inhibitor of glycogen-digesting enzyme. The insects were then stimulated to fly until they were exhausted. The flight time of each individual was recorded in the following table:

Solution injected	Samples of insect species A	Flight time (s)	Mean flight time (s)
Physiological saline	1 2 3 4 5	150 138 168 210 162	165.6
Inhibitor of trehalose-digesting enzyme	6 7 8 9 10	42 78 114 90 102	
Inhibitor of glycogen-digesting enzyme	11 12 13 14 15	132 192 174 162 156	

a) Complete the above table by calculating the mean flight time (to the nearest 1 decimal place) for the groups injected with the respective inhibitors. (1 mark)

b) With reference to the aim of the experiment, what conclusions can you draw from the data? Explain your answer. (4 marks)

c) Among individual insects, suggest ***one*** difference which led to different flight times within each group. (1 mark)

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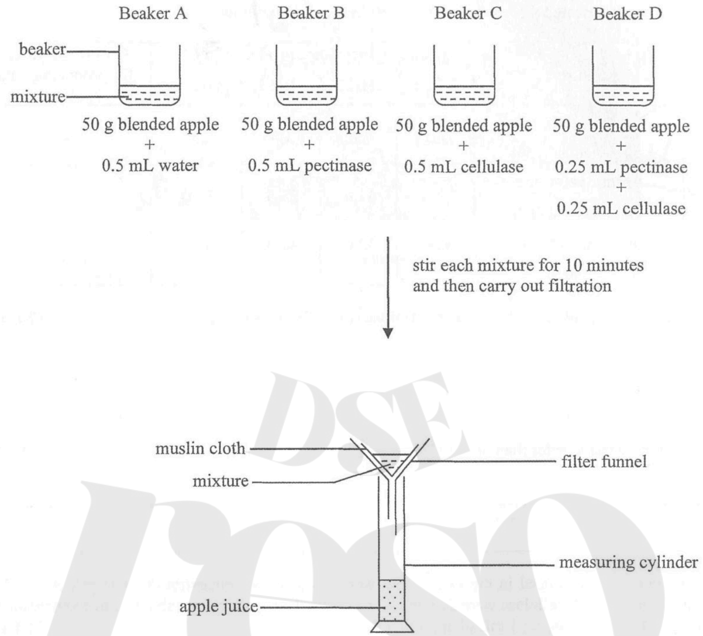
4. 2022Q5

Pectinase and cellulase are enzymes that break down the chemical components of plant cell walls. The following experiment investigates the effect of these two enzymes on the production of apple juice:



The experiment was repeated three times and the results are shown below:

Beaker	Mixture	Volume of apple juice produced (mL)				Cost of enzyme(s) for producing 1 mL apple juice
		Trial 1	Trial 2	Trial 3	Average	
A	0.5 mL water + 50 g blended apple	2.0	1.0	3.0	2.0	—
B	0.5 mL pectinase + 50 g blended apple	33.5	31.0	28.5	31.0	
C	0.5 mL cellulase + 50 g blended apple	4.5	4.0	3.5	4.0	
D.	0.25 mL pentinase + 0.25 mL cellulase + 50 g blended apple	34.0	32.0	36.0	34.0	

a) State the independent variable and dependent variable of this experiment. (2 marks)

b) Why are three trials better than one trial? (1 mark)

c) The enzyme solutions used in the experiment were at the same concentration. The prices of 0.5 mL pectinase and 0.5 mL cellulase were \$13 and \$7 respectively. Complete the above table to show the cost of enzyme(s) for producing 1 mL of apple juice. (2 marks)

d) Based on the answer in (c), which is the most cost-effective means for producing apple juice? (1 mark)

e) Explain why the apple juice collected from Beaker D is clearer than that from Beaker A. (1 mark)

Ans

2013Q6: A (47%)	2013Q7: B (44%)	2014Q9: B (70%)	2015Q3: B (60%)
2016Q27: B (18%)	2017Q5: A (60%)	2018Q2: C (54%)	2018Q3: B (76%)
2018Q4: A (69%)	2018Q27: B (96%)	2019Q9: B (72%)	2019Q10: A (22%)
2020Q6: B (70%)	2020Q7: D (41%)	2020Q8: D (78%)	2023Q15: B (60%)
2023Q16: B (61%)	2023Q17: D (57%)		

2013Q10

10 Factors determining the different conformations of protein molecules (S)

- amino acid sequence:
  - ✓ proteins are built from 20 different amino acids (1)
  - ✓ amino acid are joined together to form a polypeptide (1)
  - ✓ the amino acids sequence of the polypeptide determines the final conformation of protein molecule (1)
  - ✓ this amino acid sequence is encoded by the base sequence of a gene / code / nucleotide on DNA (1)

S = max. 3

- folding of the polypeptide:
  - ✓ the polypeptide chains then coil / fold up (1)
  - ✓ while some polypeptide chains bind together (1) to form a molecule with specific conformation

The unique shape of each protein allows different proteins to perform different functional roles in our body, e.g. it gives rise to (R)

- enzymes with unique active sites / substrate binding sites (1) that fit with specific substrates for controlling cellular metabolism (1)
- receptors with binding sites for neurotransmitters (1) that allows transmission of nerve impulse across synapse (1)
- antibodies which allow recognition of antigens / pathogens (1) for body defence (1)
- haemoglobin with binding site (1) for carrying oxygen (1)

R = max. 5

C = max. 3

11 marks

2019Q2

2 (a)
- P (1)
- because two molecules of P are joined together to form one molecule of Q (1)

(2)

(b)
- R (1)
- it has a specific site / active site for binding with P or Q (1)

(2)

4 marks

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2021Q4

4 (a)

Solution injected	Mean flight time (s)
inhibitor of trehalose-digesting enzyme	85.2
inhibitor of glycogen-digesting enzyme	163.2

(1)

(b)

Concept for mark award
• comparison of the results of the experimental set-up and control set-up to draw conclusions (1+1) x2

(4)

e.g.
- the flight time of insect species A was decreased markedly by the inhibitor of the trehalose-digesting enzyme as compared to the saline control (1), which shows that the energy supply was halted once the digestion of trehalose was inhibited; therefore, trehalose is the energy reserve for flight in insect species A (1)
- the flight time of insect species A treated with the inhibitor of the glycogen-digesting enzyme was comparable to the saline control (1), which shows that the energy supply was not affected even though the digestion of glycogen was inhibited; therefore, glycogen is not the energy reserve for flight in insect species A (1)

(c)
- the amount of trehalose storage / size of the wings / size or mass of the individuals / age of the individuals / sex (1) (accept other reasonable answers)

(1)

6 marks

2022Q5

5 (a)
- independent variable: types / combinations of enzyme(s) used (1)
- dependent variable: volume / amount of apple juice produced (1)

(2)

(b) Any ***one*** of the followings:
- increased reliability to ensure the results are reproducible (1)
- averaging the data to work out a value closer to the true value / to minimise the effect of error of experiment / individual variations (1)
- to collect data for statistical analysis (1)

(1)

(c)

Beaker	Cost of enzyme(s) for producing 1 mL apple juice
A	—
B	$ 0.42
C	$ 1.75
D	$ 0.29

(deduct 1 mark for each wrong answer)

(2)

(d)
- using a mixture of 0.25 mL pectinase and 0.25 mL cellulase / D (1)

(1)

(e)
- the enzymes break down the insoluble components of the cell wall to soluble ones (1)

(1)

7 marks